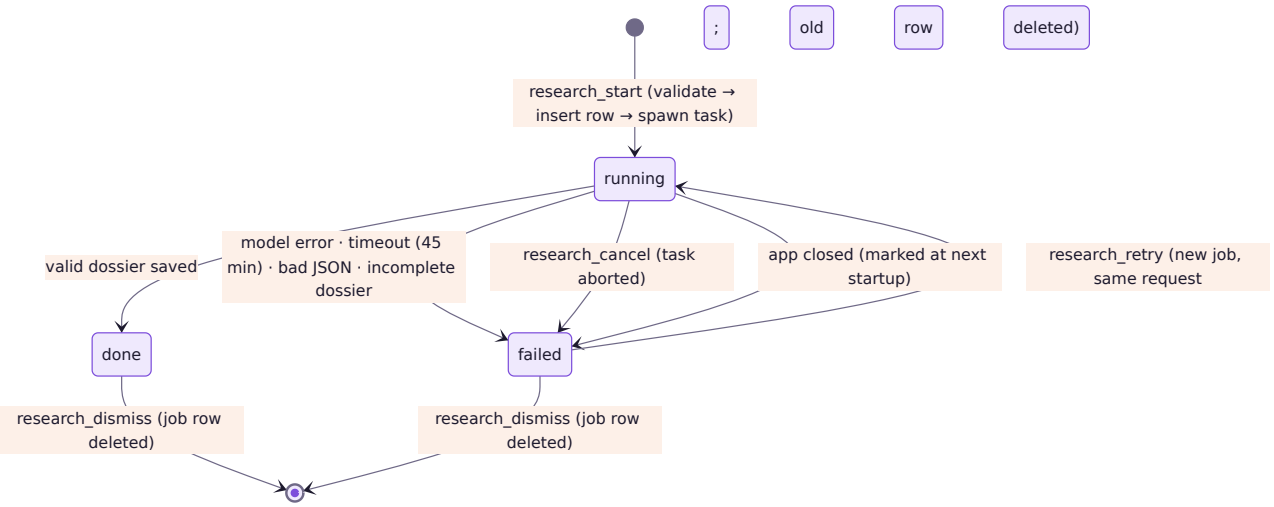
DD-04 · Research jobs and dossiers

Status: Current (v0.1.2) · **Updated:** 2026-10-30 · **Code:** `src-tauri/src/research.rs`, `dossier.rs`, `prompts/research.md`, `prompts/dossier_block.md` · **Product:** PRD-04

1. Purpose

Turn "Razorpay, SDE-2 backend" into a structured **dossier** of the real loop, from web research, without the user opening a chat window. Then feed that dossier into company-specific loops.

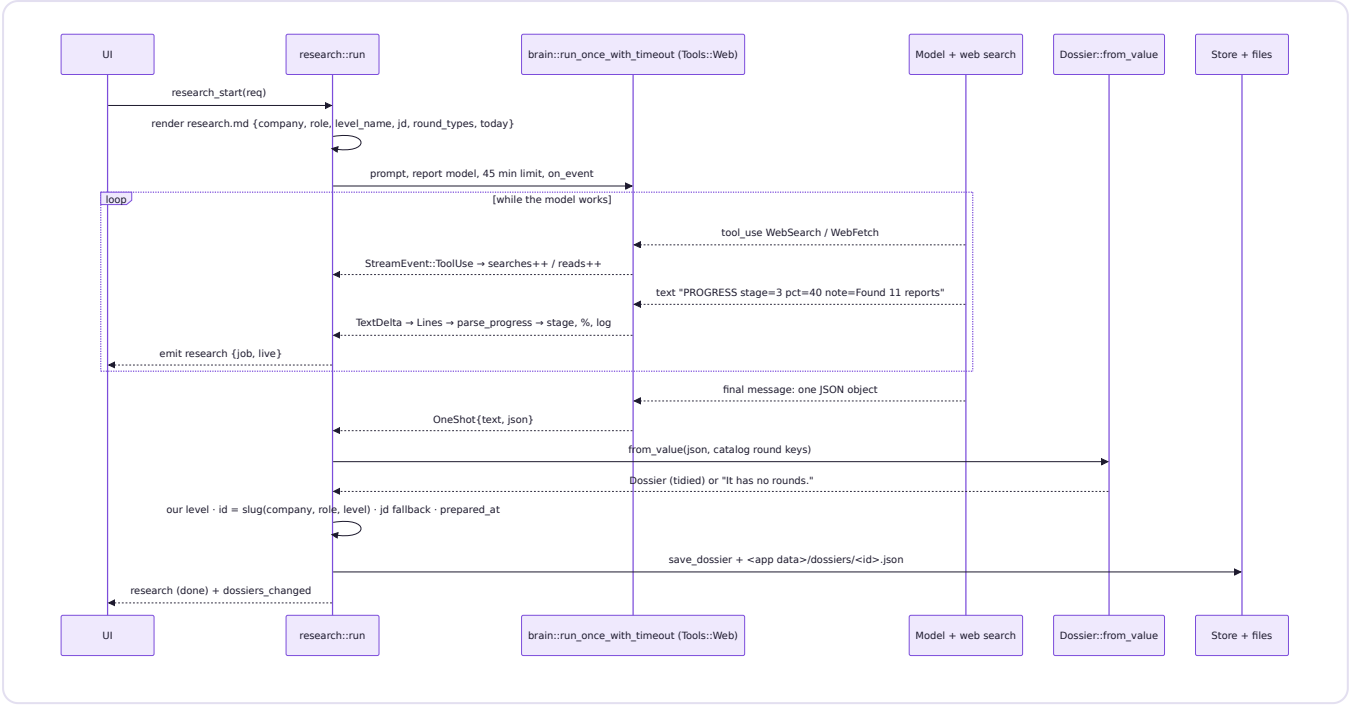
2. Job lifecycle



Piece	Kept where
Request (company, role, level, JD), status, %, log, raw output, dossier id	SQLite `research_jobs` (survives restarts)
Live counters: stage, searches, pages read, last note	Memory: `AppState.research[job_id]` (`Running { live, abort }`)
The tokio task	`AbortHandle` in the same map; cancel = abort

Every change emits a `research` event with `{job, live}`, so the card updates in real time.

3. Running a job



Progress, honestly

Two independent signals, neither faked:

1. **Tool calls** in the stream are counted as they happen: `WebSearch` → searches, `WebFetch` → pages read.
2. **PROGRESS lines** the prompt asks the model to print: `PROGRESS stage=<1-6> pct=<0-100> note=<one short sentence>`. Stage and % only ever go **up** (`max`), so a confused model can't move the bar backwards. Notes go to the log.

Stages: 1 reading request · 2 finding the current process · 3 collecting candidate experiences · 4 extracting questions per round · 5 values and interviewer style · 6 building the dossier.

Groq research isn't streamed, so only the final result arrives; Gemini counts searches from `groundingMetadata`.

4. The dossier format

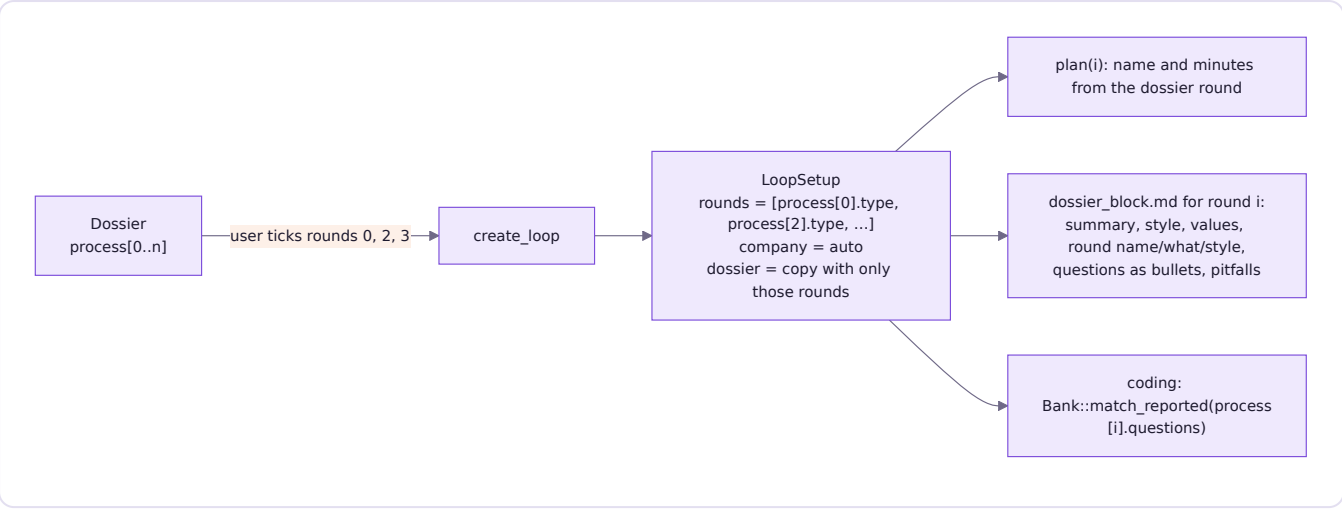
```
{
  "id": "razorpay-sde-2-backend",
  "company": "Razorpay", "role": "SDE-2 Backend", "level": "sde2",
  "preparedAt": "2026-10-27T…Z",
  "jd": "…",
  "summary": "Typical loop: DSA, machine coding/LLD, system design, HM. …",
  "style": "Practical, payments-heavy follow-ups …",
  "values": "… (source: razorpay.com/jobs)",
  "pitfalls": ["Not handling idempotency in the payment flow", "…"],
  "process": [
    {"type": "coding", "name": "Problem solving (DSA)", "minutes": 45,
     "what": "…", "style": "…",
     "questions": ["Minimum number of meeting rooms for a list of intervals", "Implement an
LRU cache; follow-up: LFU", "…"]}
  ],
  "sources": [{"title": "…", "url": "https://…"}]
}
```

The rules are in `docs/dossier.schema.json` (JSON Schema). `Dossier::from_value` applies them by hand, tolerantly:

Input problem	Tidying
`type` isn't a known round key ("Machine coding round")	`round_kind` maps by keywords in type + name: machine coding/LLD → `lld`, system design/HLD → `sysdesign`, DSA/coding → `coding`, code review/debug → `comprehension`, behaviour/values/bar raiser → `behavioral`, resume/screen/recruiter → `resume`, project/deep dive → `project`, fundamentals/core CS → `fundamentals`, else `hm`
`minutes` is `"60 min"` or missing	First number parsed, clamped to 10–180; default 45
Arrays given as strings, or strings as arrays	Normalised (`text`, `strings`)
Sources as plain URL strings	Turned into `{title:"", url}`
Missing company, role, summary, style or rounds	Rejected: "It has no summary, rounds."

Model output is **data**: it is parsed into these fields and nothing else. No field is ever run.

5. Using a dossier in a loop



The dossier block tells the interviewer to use the reported questions as the pool, to run the round the way the company does, and **never to mention having the dossier**.

6. Storage, import and export

- `save` writes the dossier to SQLite (`dossiers` table) and a pretty-printed copy to `<app data>/dossiers/<id>.json`.
- **Export** writes `<Downloads>/<id>.json`. **Import** reads any such file, validates it with the same `from_value`, and saves it ("That isn't an Onsite dossier. It has no rounds.").
- **Seed:** dossiers shipped in `dossiers/` are imported once, guarded by the `dossiers_seeded` setting.
- The slug drops the level when the role already says it: `Razorpay`, `SDE 2, backend`, `sde2` → `razorpay-sde-2-backend`.

7. Limits and ideas

Limit	Idea
A job dies with the app	Resume from the saved log, or run research in a detached process
No freshness signal	Show `preparedAt` age; offer *Refresh* after 6 months
Reported questions are free text	Link them to bank ids at save time, not at round start
Quality depends on the provider's search	Show source count and dates on the dossier card